# Adithya Hegde

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github.com/TheMonocledHamster

## Education

**Georgia Institute of Technology**, MS in Computer Science - Computing Systems — Expected May 2028

**The National Institute of Engineering**, BE in Computer Science and Engineering — Jun 2023

## Experience

**Software Engineer II**, JPMorgan Chase & Co. — Jan 2025 – Jul 2026

- Designed and delivered a new distributed data processing platform using PySpark and Java, replacing legacy systems with full ownership of architecture, implementation, and production rollout.
- Redesigned platform execution to eliminate Databricks-DynamoDB state synchronization, removing a fragile distributed dependency and improving throughput by up to 80%.
- Reduced runtime of a ∼60-operation Spark DAG processing up to 10M records from ∼10 minutes to ∼1.5 minutes (85%) through logical-plan optimization and data-dependent DAG construction.
- Tuned Spark workload partitioning and task parallelism to reduce shuffle overhead, cutting cluster compute costs by ∼40% while maintaining execution performance.
- Optimized table partitioning using Spark execution plans and Databricks query profiles, increasing data skipping from 65% to 98% and reducing dashboard refresh latency from ∼5 minutes to ∼10 seconds.

**Software Engineer I**, JPMorgan Chase & Co. — Jul 2023 – Dec 2024

- Profiled memory growth of a Polars-based transformation running in AWS Lambda's constrained runtime; introduced lazy evaluation and explicit object lifecycle control, reducing peak memory allocation by 60% and stabilizing execution latency.
- Brought up a multi-node ingestion system on Kubernetes (EKS + NiFi), implemented Airflow orchestration for distributed workflows, and operated cross-service data movement pipelines in production.
- Accelerated promotion to Software Engineer II

## Projects

**XDBC: Cross-DBMS Data Transfer | GitHub | Go, PostgreSQL** — Fall 2026

- Co-developing an end-to-end XDBC data transfer system in Go, including client/server pipelines, wire-protocol messaging, buffer management, and PostgreSQL integration.
- Reproducing XDBC data transfer benchmarks against PostgreSQL FDW baselines to evaluate throughput and transfer overhead.

## Publications

**COUNSEL - Cloud Resource Configuration Management using Deep RL** — May 2023

Proposed a Deep RL framework for cloud resource configuration using queueing-based service-chain modeling.
**Adithya Hegde**, Sameer G. Kulkarni, Abhinandan S. Prasad
10.1109/ccgrid57682.2023.00035 (**CCGRID 2023**)

## Skills

**Languages:** Python, Java, Go, SQL; Familiarity with CPP, Rust

**Data Systems:** Apache Spark, Databricks, PostgreSQL, Airflow, Pytorch

**Infrastructure:** Kubernetes, AWS, Linux